

Responses of ground beetle (Coleoptera: Carabidae) assemblages to stand characteristics and landscape structure in riparian poplar forests

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Abstract. 1. Many carabid beetles are among the most important biocontrol agents and are biological indicators of environmental change, although relatively little is known about how local and landscape factors affect forest ground beetle assemblages. Understanding such effects of multi-scale environmental drivers on ground beetles can promote carabid diversity and conservation.

2. Ground beetles were sampled by pitfall trapping in 40 poplar forest stands (sampling plots) along the Irtysh River in Northwestern China. We investigated responses of carabid activity density, species diversity and functional diversity (FD) to stand characteristics and surrounding landscape structure (the overall diversity and percentage cover of vegetation productivity classes in the surrounding area of each sampling plot based on normalised difference vegetation index).

3. While carabid activity density only responded positively to the overall diversity of vegetation productivity classes, carabid species diversity was positively associated with this landscape factor as well as vegetation cover but negatively associated with the nearest distance from sampling plot to the river. Only carabid FD responded significantly to percentage cover of vegetation productivity classes, showing a positive correlation with the proportion of medium vegetation productivity. Furthermore, overall carabid FD declined with increasing tree canopy cover.

4. Our results can help predicting spatial patterns of ground beetles according to landscape diversity and proportions of surrounding vegetation types and improve the design of conservation strategies (such as altering canopy cover) of carabids, especially forest specialist species. Any forest management strategies for promoting carabid diversity should consider the pervasive effects of landscape composition and configuration besides local stand characteristics.

Key words. Carabid beetles, functional diversity, landscape factor, riparian forests, species diversity, vegetation productivity.

Introduction

Land-use intensification and land-cover change act as the main drivers of biodiversity loss in forest landscapes while also leading to a decline in crucial ecosystem services (Straub *et al.*, 2008; Lange *et al.*, 2014). This is particularly true for riparian forest

ecosystems, which can be found throughout much of the Northern Hemisphere (Philippe & Bohlmann, 2007; Araujo Calçada *et al.*, 2015). For instance, vegetation cover in natural poplar forests along the Irtysh River in Northwest China are characterised by high spatial heterogeneity driven by enlarged farmland size and increasing grazing intensity (Lin *et al.*, 2006; Wang *et al.*, 2019). These processes are all potential causes for the loss of invertebrate biodiversity (Woodcock *et al.*, 2003; Santos *et al.*, 2019).

Conserving invertebrate biodiversity requires knowledge of effects of multiple environmental drivers at the local and

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landscape scales on arthropod groups, because most arthropod species experience their habitat at spatial scales beyond the plot level, especially natural enemies (Koivula & Vermeulen, 2005; Tschamtké *et al.*, 2007; Vele *et al.*, 2011; Lafage *et al.*, 2015). Such multi-scale effects on natural enemies mainly depend on their dispersal ability, body size, and feeding habit (Jackson & Fahrig, 2012; Philpott *et al.*, 2019). Generalist predators are known to strongly respond to factors at the landscape scale owing to high dispersal and colonisation capacity, while small and specialised natural enemies like egg parasitoids are always influenced by local factors (Chaplin-Kramer *et al.*, 2011). Many ground beetles, which are generalist predators, have been widely recommended as bioindicators of environmental change at different spatial scales (Rainio & Niemelä, 2003; Woodcock *et al.*, 2003, 2010; Alalouni *et al.*, 2013; Borchard *et al.*, 2014). Understanding the mechanisms that maintain fauna diversity and the associated services are key to improve effectiveness of conservation actions (Anton *et al.*, 2010), especially given the current scenario of global changes (Dawson *et al.*, 2011).

Carabid abundance, species diversity and functional diversity (FD) are likely to influence the provision of ecosystem services, and these broad characteristics can be used to assess the multi-scale drivers of ground beetle assemblages (Woodcock *et al.*, 2010; Spake *et al.*, 2016; Philpott *et al.*, 2019). Study on the effects of environmental factors on carabid biodiversity has traditionally utilised a taxonomic approach by focusing on their abundance or species richness (Batáry *et al.*, 2008; Fountain-Jones *et al.*, 2015). But functional-trait analysis has several advantages over taxonomic approach in predicting various ecosystem functions and services (Kleyer *et al.*, 2012; Fountain-Jones *et al.*, 2015; Magura, 2017). FD captures the relative value, range or dispersion of functional traits in a community (Petchey & Gaston, 2002, 2006; Woodcock *et al.*, 2014; Gallé *et al.*, 2019). Thus, it is increasingly used as a metric to evaluate the responses of ground beetles to environmental change (Pavoine & Bonsall, 2011; Spake *et al.*, 2016).

Ground beetles in meadow, urban or agricultural ecosystems have been proved to be influenced by both local and landscape factors (Woodcock *et al.*, 2010; Lafage *et al.*, 2015; Füsser *et al.*, 2017; Philpott *et al.*, 2019), and most studies of carabid diversity in forest ecosystems focused on effects of local forest management (Lange *et al.*, 2014; Negro *et al.*, 2014), forest stand types (Barsoum *et al.*, 2014), or stand characteristics (Spake *et al.*, 2016). However, relatively little is known about the combined and possible interactive effects of local stand factors and landscape structure of surrounding vegetation on forest ground beetle assemblages to our knowledge. As ground beetles have experienced general declines in diversity and abundance across natural forests in the northwest China, recognising the critical local and landscape-level factors affecting their abundance and diversity is important to develop comprehensive management strategies for promoting carabid diversity and conservation (Lafage *et al.*, 2014, 2015; Rahman *et al.*, 2015; Philpott *et al.*, 2019).

In order to quantify the landscape structure of surrounding vegetation, landscape metrics is often derived from the classification of land-cover, which consists of a few discrete classes (Macfadyen *et al.*, 2015). However, plant species in natural

poplar forests could not fall neatly into certain habitat-type classes due to the high spatial heterogeneity of vegetation resources, where land cover is dominated by woodlands, grasslands, shrublands and farmlands (Vasseur *et al.*, 2013). Gross primary productivity (GPP, hereafter vegetation productivity) is commonly used to characterise carbon-fixing ability of plants through photosynthesis in terrestrial ecosystems, which can mediate the relationship between land cover change and biodiversity (Running *et al.*, 2004; Weiskittel *et al.*, 2011; Alberti, 2015; LeVine & Crews, 2019). Therefore, in this study, landscape variables were generated from different vegetation productivity classes based on remotely sensed data (Lafage *et al.*, 2014) in order to depict the spatial distribution of forest vegetation at the landscape scale and assess how this impacts carabid assemblages among different forest stands.

Here, we investigate how ground beetle activity density, species diversity and FD respond to stand characteristics (e.g. vegetation cover, tree canopy cover and the nearest distance from sampling plot to the river) and landscape structure of surrounding vegetation in 40 forest stands growing in natural poplar forests along the Irtysh River in Northwestern China. We used the overall diversity and percentage cover of vegetation productivity classes in buffers of 300 m radius surrounding all sampling plots as proxies for landscape structure of surrounding vegetation. Specifically, we hypothesised that (i) activity density of ground beetles would respond primarily to overall diversity of vegetation types in surrounding landscape, as many of them are highly mobile generalist predators and thus would benefit from different prey resources within forest landscapes; (ii) species diversity of ground beetles would also be higher in forest landscapes with higher diversity of vegetation types based on the 'habitat heterogeneity hypothesis' (Tews *et al.*, 2004); and (iii) ground beetle FD would show a greater dependence on landscape structure of surrounding vegetation than local stand factors, a response that is attributed to their high mobility needed for movement between dynamically variable habitat types (Woodcock *et al.*, 2010).

Materials and methods

Study area and sampling plots

The study was conducted in natural poplar forests along the Irtysh River in Altay city, Xinjiang Uygur Autonomous Region, Northwest China (87°47'40" to 87°53'21" E; 47°20'48" to 47°22'03" N) (Fig. 1a). Our study area is mainly located in a forest mosaic landscape comprised by woodlands, grasslands, farmlands, and bare ground during the vegetation growing season on the south bank of this river. With a temperate continental climate, natural tree species growing widely on the banks of the Irtysh River mainly include *Populus laurifolia* Ledeb., *Populus alba* L., and *Populus nigra* L., and such riparian forests are regarded as an important gene pool for plants in the family Salicaceae. But natural riparian forests have been largely converted into farmlands or have become the subject of increasing grazing intensification, and the special environmental characteristics (such as drought and high temperature in the growing season)

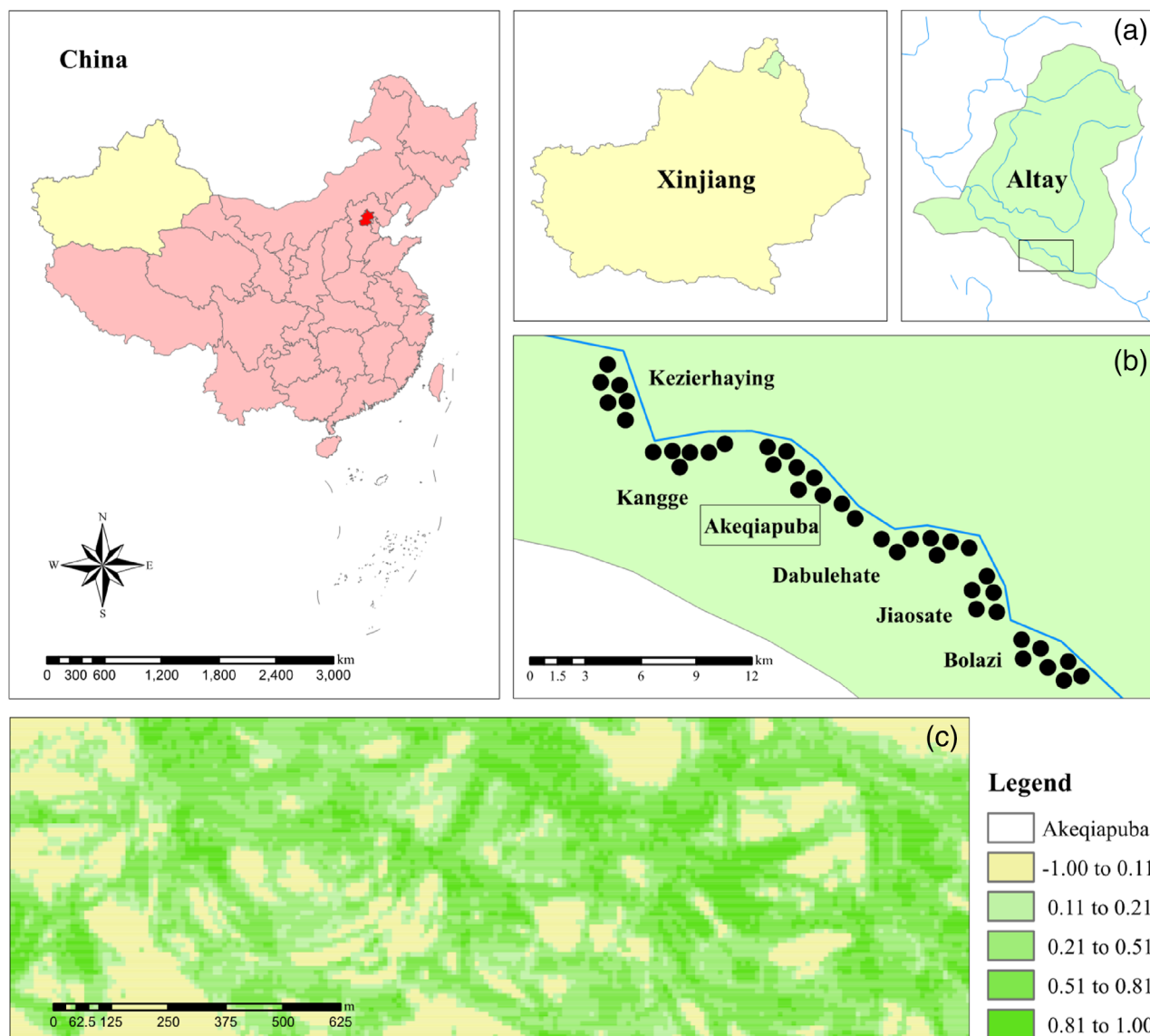


Fig 1. (a) Location of the study area in Altay city, Xinjiang Uygur Autonomous Region, Northwest China; (b) Distribution of 40 sampling plots at different study sites in six villages: seven plots in Bolazi village; five plots in Jiaosate village; seven plots in Dabulehate village; nine plots in Akeqiapuba village; six plots in Kangge village; six plots in Kezierhaying village (black circles were sampling plots within a 300 m radius buffer at which we quantified landscape variables); (c) Example of a village (Akeqiapuba) where we classified vegetation productivity into five classes. [Color figure can be viewed at wileyonlinelibrary.com]

in the river basin area often cause stress on growth and development, defence and resistance to insect pests of poplars.

In the study area, no physical and chemical means were applied to *Populus* species for the purpose of pest control within natural riparian forests. Sampling plots were distributed unevenly at different study sites in six villages: seven plots in Bolazi village; five plots in Jiaosate village; seven plots in Dabulehate village; nine plots in Akeqiapuba village; six plots in Kangge village; and six plots in Kezierhaying village (Fig. 1b). Thus, a total of 40 sampling plots of 40 m × 40 m were selected by a stratified random design in each of the six study sites along a vegetation cover gradient.

Distance between the centres of each sampling plots was at least 600 m to reduce spatial autocorrelation. Each of the 40 sampling plots was located in poplar stand dominated by *Populus laurifolia*, which was the main tree species in all study sites.

Ground beetle sampling

In each sampling plot, ground beetles were sampled by continuous pitfall trapping. As beetle catches in pitfall traps reflect the activity of the species, trap catches indicate ground beetle

activity density, not necessarily abundance (Niemelä & Kotze, 2009; Philpott *et al.*, 2019). On the diagonal of each sampling plot, five traps were installed 10 m apart on a transect that is parallel to the southern bank of river, and traps were almost level with each other. We inserted plastic cups (diameter = 6.5 cm, depth = 9.5 cm) into the ground by using a soil auger and ensured that the top surface of each pitfall trap flushed with the soil surface. We then used 120 ml of 50% ethylene glycol and water solution as a temporary preservative and a few drops of colourless, odourless detergent to reduce the surface tension (Gallé *et al.*, 2019). Pitfall trapping was carried out over a 16-week period during the growing season from mid-May to mid-September for two consecutive years (2018 and 2019), because this period is generally associated with a maximum abundance of these arthropods. Pitfall traps were emptied fortnightly, for a total of eight times before the periodical mowing of the above-ground plants by local herdsman. We transferred all samples into 70% ethanol for further identification. Adult ground beetles were identified to the species level by expert entomologists according to Berlov (2002), Dudko *et al.* (2010), and Kataev and Liang (2015) in the laboratory. All carabid specimens are stored at the Museum of Beijing Forestry University.

Functional diversity

Three ecological traits that characterised functional diversity between ground beetle species included: body size (large vs. small), feeding trait (carnivorous vs. omnivorous), and wing morphology (macropterous vs. brachypterous) (Homburg *et al.*, 2014) (Supporting Information Table S1). Body length was used as a measure of body size, which was measured as the distance from the tip of the mandibles to the end of the abdomen. Carabid species with an average body size equal to or longer than 15 mm were considered large; otherwise, they were considered small (Gobbi *et al.*, 2007). Carabid species were considered carnivorous if they were strictly predators during their life cycle, whereas species that were occasional predators or herbivores were considered omnivorous (Gobbi & Fontaneto, 2008). For wing morphology, macropterous species were those with wings longer than the elytra, whereas brachypterous species were those with wings shorter than the elytra (Brambilla & Gobbi, 2014). Carabid FD was measured by calculating Rao's quadratic entropy (FD_Q), which is a multi-trait FD metric that describes the diversity of a given community, combining the relative abundances of species with the biological information about the functional differences between species (Rao, 1982; Ricotta, 2005). FD_Q expresses the average functional dissimilarity between two randomly selected individuals if relative abundance is calculated from the number of individuals, and it has been widespread used to characterise human impacts on ecosystem processes (Aubin *et al.*, 2013). We calculated FD_Q for each sampling plot as:

$$FD_Q = \sum_{i=1}^S \sum_{j=1}^S d_{ij} p_i p_j,$$

where S is the number of species in a sampling plot, d_{ij} is the dissimilarity in trait values between each species pair i and j , and

p is the proportion of individual number per species relative to the total individual number within a plot. Rao's quadratic entropy was calculated by using the R package FD.

Landscape structure of surrounding vegetation

Landscape structure of surrounding vegetation was measured in a 300 m radius buffer (Wang *et al.*, 2019). The geographical coordinates of the centre of each sampling plot were collected using a handheld GPS unit. Then, based on these coordinates, open-access remote sensing images were acquired on 28 June 2018 for the whole study area covering six villages from the European 'Sentinel-2A' satellite. We preprocessed remote sensing images using ENVI version 5.4 software in order to eliminate errors and improve their quality, including radiation correction and atmospheric correction. The results of image-interpretation were rectified by adequate ground-truth data recorded in a forest vegetation survey. The normalised difference vegetation index (NDVI) is often used as a preferred surrogate for vegetation productivity of the terrestrial ecosystems (Voepel *et al.*, 2011; Lezama *et al.*, 2014), particularly in applications involving land-cover change. Vegetation productivity was thus quantified by calculating NDVI values, which is defined as: $NDVI = (R_{NIR} - R_{red}) / (R_{NIR} + R_{red})$, where R_{NIR} and R_{red} refer to the reflectance values derived from spectral radiances measured by the near-infrared channel and the red visible channel, respectively. NDVI values range from -1 (water) to 1 (maximum vegetation signal). We calculated NDVI values for each of 300 m radius buffers around sampling plots using 'Band Math' tool in ENVI version 5.4 software applied to the remote sensing image and classified NDVI values into five vegetation classes representing the intensity of the vegetation signal: (i) low vegetation (-1.00 to 0.11); (ii) medium-low vegetation (0.11–0.21); (iii) medium vegetation (0.21–0.51); (iv) medium-high vegetation (0.51–0.81); and (v) high vegetation signal (0.81–1.00) (Fig. 1c). We distinguished each vegetation class based on a variation range examination of how NDVI values change in each habitat type (e.g. bare ground, shrubland, grassland, woodland, or farmland) across time. These vegetation productivity classes defined different habitat patches that had similar vegetation growth characteristics, for instance, NDVI values below 0.2 typically represent very low vegetation signal or bare ground. Using five classes enabled us to separate the amount of each vegetation productivity class during the growing season precisely (Macfadyen *et al.*, 2015). Finally, the proportion of each of the five vegetation productivity classes was calculated in a buffer of 300 m radius centred on each sampling plot. Shannon's Diversity Index (SHDI) was also calculated by using Fragstats version 4.2 to quantify the overall diversity of vegetation productivity classes.

Stand characteristics

In each sampling plot, forest and vegetation structure was described by measuring a series of stand characteristics. Within the core area of each plot, diameter at 130 cm height [diameter

at breast height (DBH)] and tree height of all living trees with DBH > 5.0 cm were recorded. As DBH is often used as a proxy of tree height in most tree species, we only considered DBH as measure of tree size, which is an important structural trait, critical in forest ecology and for above ground biomass estimate. Canopy cover was assessed at four continuous 10 m × 10 m assessment areas along the transect using a visual cover method (Humphrey *et al.*, 1999). Percentage of vegetation cover on each plot was calculated from 10 random 1 m × 1 m quadrats along the transect (Rahman *et al.*, 2015). We also measured species richness and average height of herbaceous plant species in three random 5 m × 5 m sampling subplots within each sampling plot. An important environmental factor, the nearest distance from the centre of sampling plot to river bank (the distance to river, Magagula, 2003; Rykken *et al.*, 2007) was measured by using ENVI version 5.4 based on satellite images.

Statistical analysis

At each sampling plot, the activity density and species richness data of ground beetles from the five pitfall traps on each transect and for the two sampling years were pooled for statistical analyses. The responses of carabid species diversity and FD to landscape structure of surrounding vegetation and stand characteristics were assessed using linear mixed-effect models (LMMs) for each response variable. The 'lmer' function (R package lme4; Bates *et al.*, 2015) were used for species diversity of ground beetles, represented by the Shannon-Wiener's diversity index (H') (function 'diversity' in R package vegan; Oksanen *et al.*, 2019) and FD of ground beetles (FD_Q) as they are continuous responses. We tested the effects of stand level and landscape factors on carabid activity density using generalised linear mixed-effect models (GLMMs) with a negative binomial distribution (function 'glmer.nb' in R package lme4; Bates *et al.*, 2015) since overdispersion was detected (overdispersion test via R package qcc; Scrucca, 2014). For variable selection, we conducted Spearman's correlation test to check for correlations among the proportions of five vegetation productivity classes, the SHDI and all stand characteristics. There was a strong correlation between the DBH and canopy cover, and the latter has previously been observed to affect ground beetle assemblages in forests (Lange *et al.*, 2014; Negro *et al.*, 2014; Spake *et al.*, 2016), thus DBH was not considered to be an explanatory variable for further analyses. Additionally, as species richness and average height of herbaceous plant species, the proportions of low and medium-low vegetation productivity showed a strong correlation (correlation coefficients > |0.7|) with other explanatory variables (Supporting Information Table S2), we excluded them from further analyses (Yang *et al.*, 2019). Variance inflation factor (VIF) values were calculated to assess collinearity in the remaining seven explanatory variables and were found to be <2.0, indicating that covariation between explanatory variables was not a problem (Dormann *et al.*, 2013; Schirmel *et al.*, 2018). Therefore, explanatory variables in all full models were SHDI, the proportions of medium vegetation productivity (Medium VP), medium-high vegetation productivity (Medium-high VP), high vegetation productivity (High VP), canopy cover,

vegetation cover, and the distance to river. All explanatory variables were assigned as fixed effects, and village ID was included as a random effect in order to account for the non-independence of sampling plots within the same village. Furthermore, explanatory variables in all models were scale-centred to facilitate the model's convergence and two-way interactions between landscape and stand variables were tested only, as their three-way interactions could not be explored further owing to the relatively large number of explanatory variables (Zuur *et al.*, 2009).

For model simplification, we used an information-theoretic approach to identify the best set of models from global GLMMs via Akaike's Information Criterion corrected for small sample size (AICc) (Grueber *et al.*, 2011). All possible additive models were constructed using maximum likelihood methods in package MuMIn (Bartoń, 2018). The input variables were standardised using Gelman and Su's (2018) approach (function 'standardise' in R package arm), which was essential for interpreting the parameter estimates after model averaging. We applied full model averaging across all plausible models with <2 $\Delta AICc$ of the best-fit model (i.e., the model with the lowest AICc value) (Grueber *et al.*, 2011; Schirmel *et al.*, 2018). We calculated marginal R^2 (R_m^2) and conditional R^2 (R_c^2) to assess the proportion of variance explained by fixed effects and by both the fixed and random effects and thus estimate the goodness-of-fit of all plausible models following Nakagawa and Schielzeth (2013). We also conducted Moran's I test for the residuals of all models to test for spatial autocorrelation (function 'moran.test' in R package spdep) (Bivand, 2018) and found no evidence of spatial autocorrelation. All statistical analyses were performed in R version 3.5.1 (R Core Development Team, 2018).

Results

A total of 5503 ground beetles, belonging to 16 species, were collected during the two-year sampling period. The most abundant carabid species were *Amara curta*, *Poecilus cupreus*, *Anchomenus dorsalis*, *Anisodactylus binotatus*, *Agonum impressum* and *Poecilus punctulatus* in forest landscapes, representing 70.4% of the total carabid catches. The majority of collected ground beetles were small and medium-bodied predators (68.5% of individuals), and overall 74.5% of the total individuals were capable of flying (Supporting Information Table S1).

Effects of environmental factors on activity density of ground beetles

Results of GLMMs and LMMs showed that ground beetle activity density, species diversity and FD responded differently to landscape structure and stand characteristics, but explanatory variables had no significant interaction effects (Supporting Information Table S3).

Activity density of ground beetles varied among different poplar forest stands (36–817 individuals per plot over the sampling period). The results of multi-model inference showed that the model 'SHDI + Dis. to river' was the best ranked model

(Supporting Information Table S3). The summarised results of the model averaging showed that landscape structure of surrounding vegetation had the strongest effect on carabid activity density, but stand characteristics had no significant effect on it (Table 1). More specifically, carabid activity density significantly increased with increasing the overall diversity of vegetation productivity classes (Estimate = 0.219, $P < 0.001$) (Fig. 2a). The value for the variance that is taken by the random terms was 0.017.

Effects of environmental factors on species diversity of ground beetles

Species diversity of ground beetles varied among different poplar forest stands (0.43–2.00 per plot over the sampling period). It responded strongly to both landscape structure of surrounding vegetation and stand characteristics (Table 1). The model 'SHDI + Medium VP + High VP + Vegetation cover + Dis. to river'

was the best ranked model for species diversity of ground beetles, (Supporting Information Table S3). While the species diversity of ground beetles responded positively to the overall diversity of vegetation productivity classes (Estimate = 0.671, $P < 0.001$) (Fig. 2b), the proportion of each vegetation productivity class was not found to have a significant effect on carabid species diversity. Furthermore, carabid species diversity was positively associated with vegetation cover (Estimate = 0.286, $P < 0.001$) (Fig. 3a) but negatively associated with the nearest distance from sampling plot to the river (Estimate = -0.148, $P < 0.001$) (Fig. 3b). The value for the variance that is taken by the random terms was 0.045.

Effects of environmental factors on functional diversity of ground beetles

Landscape structure of surrounding vegetation and stand characteristics together had strong effects on FD of ground beetles, and the best ranked model of carabid FD was 'Medium VP + Medium-

Table 1. Summary results of the model averaging to assess responses of activity density, species diversity and functional diversity of ground beetles to the overall diversity of vegetation productivity classes (SHDI), the proportions of medium vegetation productivity (Medium VP), medium-high vegetation productivity (Medium-high VP), high vegetation productivity (High VP), canopy cover, vegetation cover and the nearest distance from sampling plot to river (Dis. to river).

Response variables		Explanatory variables	Estimate	Std. Error	z-value	P-value	Relative importance
Activity density	Fixed effects	Intercept	2.724	0.183	24.194	<0.001	
		SHDI	0.219	0.186	1.135	<0.001	1.00
		Medium VP	-0.438	0.191	2.218	0.265	0.25
		Canopy cover	-1.224	0.169	6.979	0.257	0.26
		Dis. to river	-0.193	0.045	4.169	0.655	0.23
		Medium VP:Canopy cover	0.186	0.118	1.517	0.129	0.25
		Medium VP:Dis. to river	1.296	0.113	2.919	0.351	0.22
	Radom effects	Groups	Name	Variance	Std. Dev.		
		Study sites	Intercept	0.017	0.130		
Species diversity	Fixed effects	Intercept	4.789	0.090	51.713	<0.001	
		SHDI	0.671	0.081	8.009	<0.001	1.00
		Medium VP	0.159	0.050	3.048	0.230	0.45
		Canopy cover	-0.131	0.039	3.245	0.117	0.58
		Vegetation cover	0.286	0.033	6.981	<0.001	1.00
		Dis. to river	-0.148	0.037	3.845	<0.001	1.00
	Radom effects	High VP	0.288	0.053	5.260	0.147	0.32
		Medium VP: Vegetation cover	0.124	0.686	1.74	0.082	0.55
		Medium VP: Dis. to river	-0.228	0.101	2.179	0.293	0.22
		Groups	Name	Variance	Std. Dev.		
Functional diversity	Fixed effects	Study sites	Intercept	0.045	0.211		
		Intercept	4.734	0.086	52.731	<0.001	
		Medium VP	0.170	0.056	2.936	0.003	1.00
		Medium-high VP	0.213	0.070	2.947	0.321	0.59
		High VP	0.577	0.085	6.561	0.160	0.68
		Canopy cover	-0.162	0.047	1.290	0.019	1.00
	Radom effects	Vegetation cover	-0.305	0.037	7.839	0.068	0.76
		Dis. to river	-0.393	0.045	8.405	0.306	0.55
		Medium VP: Vegetation cover	-0.018	0.047	0.380	0.704	0.34
		Medium VP: Dis. to river	-1.249	0.067	1.234	0.217	0.29
	Radom effects	Groups	Name	Variance	Std. Dev.		
		Study sites	Intercept	0.029	0.169		

Significant values ($P < 0.05$) are highlighted in bold; relative importance is the sum of the Akaike weight associated with each explanatory variable in the top model set; the variance and the standard deviation (Std. Dev.) of the random effects are also presented for each model.

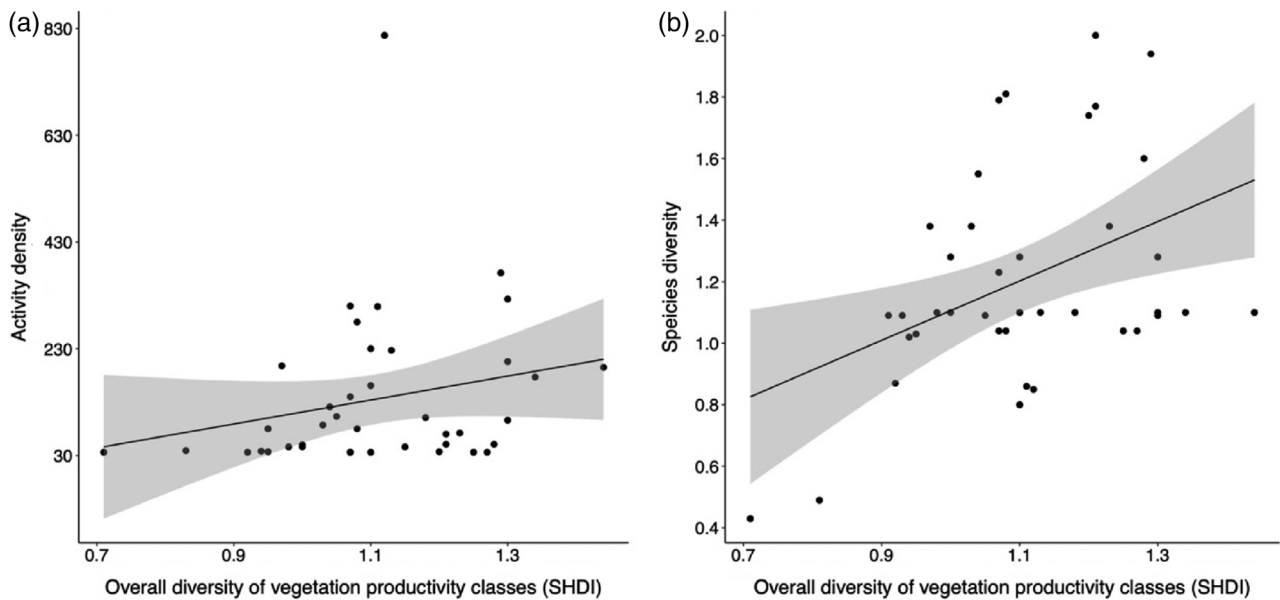


Fig 2. The responses of (a) activity density and (b) species diversity of ground beetles to the overall diversity of vegetation productivity classes (SHDI). Regression lines and 95% confidence intervals are shown. The shaded area around regression lines represent glmer/lmer prediction confidence intervals.

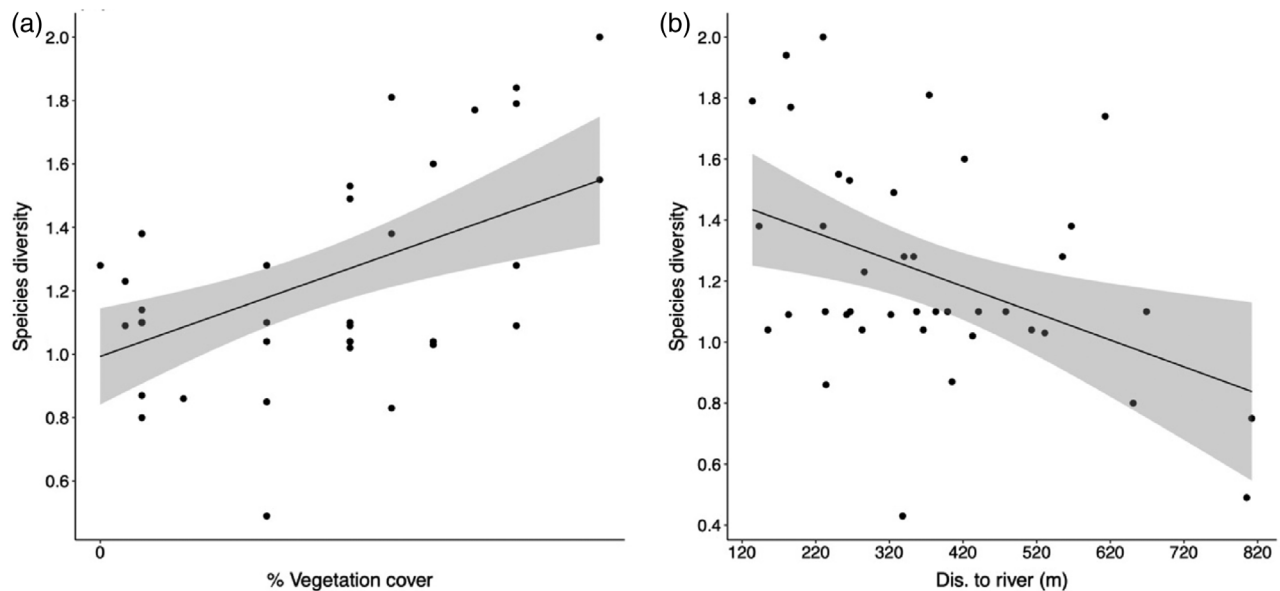


Fig 3. The responses of species diversity of ground beetles to (a) vegetation cover and (b) the nearest distance from sampling plot to river (Dis. to river). Regression lines and 95% confidence intervals are shown. The shaded area around regression lines represent lmer prediction confidence intervals.

High VP + High VP + Canopy cover + Vegetation cover + Dis. to river' (Supporting Information Table S3). The multi-model inference of LMMs showed that FD of ground beetles significantly increased with the increasing proportion of medium vegetation productivity (Estimate = 0.170, $P = 0.003$) (Fig. 4a) in surrounding landscape; however, the effect of overall diversity of vegetation classes on carabid FD was not significant (Table 1). Moreover, we found a negative effect of tree canopy cover on carabid FD

(Estimate = -0.162 , $P = 0.019$) (Fig. 4b). The value for the variance that is taken by the random terms was 0.029.

Discussion

Our study revealed that landscape structure and stand characteristics had different effects on activity density, species diversity

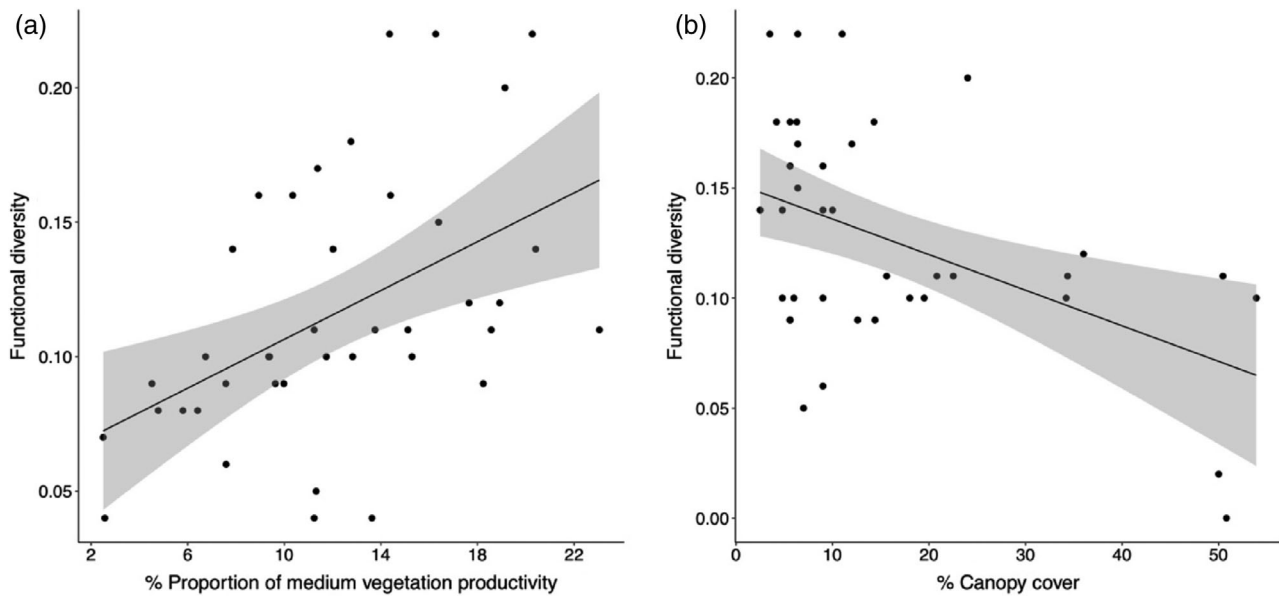


Fig 4. The responses of functional diversity of ground beetles to (a) the proportion of medium vegetation productivity and (b) tree canopy cover. Regression lines and 95% confidence intervals are shown. The shaded area around regression lines represent lmer prediction confidence intervals.

and FD of ground beetles in natural riparian poplar forests. In particular, all three broad characteristics of ground beetles correlated with landscape structure of surrounding vegetation, which suggested that spatial distribution of vegetation productivity is a good proxy for landscape structure explaining species composition and distribution patterns of ground beetle assemblages in forest ecosystems (LeVine & Crews, 2019). Although habitats in riparian border within natural poplar forests had different vegetation cover and soil moisture (Wang *et al.*, 2019), the landscape classification map of riparian habitats by using traditional methods of classification could not result in a good explanation because it was difficult to classify mosaic-pattern vegetation into certain land-cover class neatly (Vasseur *et al.*, 2013). Our findings are thus based on the classification of five vegetation productivity classes and these findings might help improve landscape-scale forest management strategies for conserving ground beetle diversity (Weiskittel *et al.*, 2011; Lange *et al.*, 2014; Macfadyen *et al.*, 2015).

Effects of landscape diversity on carabid activity density

Significant positive effect of the overall diversity of vegetation productivity classes on carabid activity density did support our first hypothesis that activity density of ground beetles would respond primarily to landscape structure of surrounding vegetation. Vele *et al.* (2011) investigating the effects of local site conditions and landscape factors on ant and carabid community composition in a spruce forest, found that a greater quantity of suitable habitat in the surrounding landscape increased the probability of greater abundance of carabids in both optimal and sub-optimal patches, similar to our result. Philpott *et al.* (2019) also demonstrated that landscape factors (gardens with higher crop

richness and that are surrounded by agricultural land) influenced carabid activity density. Many ground beetles are highly mobile generalist predators, they can inhabit a variety of habitats that differ in their food resources, especially in terms of prey species and abundance (Woodcock *et al.*, 2010). The great reliance on multiple prey resources over their lifetimes can increase the dependence of ground beetles on landscape structure of the surrounding vegetation (Purtauf *et al.*, 2005; Fusser *et al.*, 2017). As the diversity of habitats within a landscape increases, so too will variability in the abundance of potential prey items in each landscape habitat parcel (Rooney *et al.*, 2008). Therefore, the overall diversity of different vegetation classes in the landscape can enhance their movement between prey resources within different habitats and thus increase the total activity density of ground beetles within a forest landscape, which was shown to be the case in this study.

Effects of landscape diversity and stand factors on carabid species diversity

In accordance with our second hypothesis, we found that carabid species diversity was positively correlated with overall diversity of vegetation productivity classes. The 'habitat heterogeneity hypothesis' is one of the most important basis of landscape ecology. Habitat heterogeneity is often associated with higher biodiversity in ecosystems (Benton *et al.*, 2003). It is considered that habitats with higher spatial heterogeneity in vegetation contain more microclimatic conditions and ecological niches to meet survival needs of more arthropod species and thus increase their species diversity on multiple spatial scales (Davidowitz & Rosenzweig, 1998). Tews *et al.* (2004) reviewed studies on the relationship between habitat heterogeneity/

diversity and animal species diversity in a selected literature survey and found that majority of studies showed a positive correlation. Increasing landscape habitat diversity is generally associated with increase in species diversity of generalist predators (Gardiner *et al.*, 2009; Chaplin-Kramer *et al.*, 2011). This was also found to be the case for species diversity of ground beetles that showed a positive correlation with landscape habitat diversity. Many ground beetles show a preference for more than one potential habitat and their prey are not restricted to one particular habitat (Barsoum *et al.*, 2014; Fusser *et al.*, 2017). As habitat diversity increases, so too does the chance of encountering ground beetle species associated with these different habitat types at a particular sampling location. For mobile carabid species, heterogeneous landscapes with a greater range of resources and microclimates can generate more stable population dynamics (Oiver *et al.*, 2010), so carabid species diversity was associated with higher habitat diversity of surrounding landscape (Taboada *et al.*, 2006; Balzan *et al.*, 2016).

Additionally, species diversity of ground beetles responded strongly to stand characteristics, supporting the results of a research aimed to examine the influence of natural areas on carabid communities, which found that environmental factors at both local and landscape scales influenced ground beetle diversity and community composition (Werling & Gratton, 2008). Vele *et al.* (2011) also indicated that carabid community composition was strongly affected by both local conditions near the pitfall traps and the availability of optimal habitat in the surrounding landscape. In this study, at the local scale, vegetation cover and the nearest distance from sampling plot to the river were found to be the two most influential stand characteristics for carabid species diversity. Vegetation cover and distance from running/standing water are known to be the main determinants of carabid beetle fauna (Magagula, 2003). Several previous studies suggested that species diversity of carabid beetles had a strong relationship with local habitat factors, and vegetation cover was found to be the most influential local factor for increasing carabid species diversity (Niemelä *et al.*, 2002; Rahman *et al.*, 2015). Vegetation cover can modify microclimatic conditions and provide ideal living space to promote the community establishment of carabid beetles and thus increase diversity of different carabid species (Rahman *et al.*, 2015). Schlegel & Riesen (2012) also showed that high vegetation cover turned out to be the crucial habitat factor for carabid species composition and diversity. Furthermore, riparian border, a habitat with higher vegetation cover, was shown to have the highest species diversity of carabid species (Magagula, 2003). In riparian border, streams can change microclimate environment such as humidity and temperature of the adjacent terrestrial habitat, resulting in diverse animal communities (Lavallee & Richardson, 2010). A study on terrestrial ground beetle communities suggested that riparian habitats had a great influence in maintaining their species diversity in forests, in part because the changing microclimate in riparian border is the determining factor in habitat choices of different ground beetles (Ryken *et al.*, 2007). Therefore, in natural poplar forest ecosystems, the higher vegetation cover can support more ground beetle species that rely on diverse microclimates of riparian habitats. This study demonstrated the potential importance of vegetation cover and the distance to the river to carabid species diversity.

Effects of canopy cover and landscape structure on carabid functional diversity

We found that local and landscape-level factors all strongly affected carabid FD, which is inconsistent with our third hypothesis that FD of ground beetles would show a greater dependence on landscape structure of surrounding vegetation than local stand factors. However, this finding is in accordance with Gallé *et al.* (2019), who demonstrated that landscape configuration, management method and plot position drive carabid FD. Our results indicated that carabid FD decreased with increasing tree canopy cover, which was the most important stand factor at the local scale. For ground beetles, open-habitat species have different functional traits from forest specialist species especially in hunting strategies and dispersal abilities (Fuller *et al.*, 2008; Jelaska & Durbešić, 2009). Lower percentages of canopy cover generally cause a lower humidity and a higher temperature but can also promote the growth of grasses and shrubs, which are important habitats for open-habitat carabids, providing ample space to colonise, disperse, and hunt (Negro *et al.*, 2014). It means that the more open forest structure and the more diverse vegetation on the forest floor may promote immigration of additional open-habitat carabid species (Lange *et al.*, 2014). Therefore, ground beetles may easily re-establish as the forest canopy becomes less continuous caused by tree mortality or enlarged farmland size in this study. Our finding of decreasing carabid FD with increased canopy cover can be attributed to a deduction of open-habitat species in forest stands with higher canopy cover (Spake *et al.*, 2016). In addition, the positive relationship between carabid FD and the proportion of medium vegetation productivity could be due to a fact that the medium-class vegetation productivity in this study was mainly associated with the percentage of grasslands in the surrounding landscape. At the landscape level, the proportion of open vegetation patches, such as non-forest vegetation areas, within 300 m of a trap appears to be an important factor for ground beetles (Vele *et al.*, 2011). Grasslands in the landscape are important habitats for open-habitat species in forest ecosystems and support a wide variety of ground beetle species (Woodcock *et al.*, 2005; Rahman *et al.*, 2015). Successful migration of ground beetles is a key aspect for their survival, which indicates that the movement of open-habitat species can be promoted by the presence of grasslands nearby (Bodin & Norberg, 2007; Jelaska & Durbešić, 2009). Due to the limited dispersal ability of many forest specialist species, colonisation of non-forest vegetation areas by these species depends on landscape elements such as the proximity to source habitats and matrix quality (Koivula *et al.*, 2002). In contrast to open-habitat species, forest species are reluctant to cross open habitat (Fuller *et al.*, 2008). As FD of ground beetles strongly responded to landscape structure of surround vegetation, the great complementarity in functional traits associated with different hunting strategies and dispersal ability would be likely to sustain functionally diverse ground beetle communities and potential biocontrol at a large spatial scale (Straub *et al.*, 2008; Woodcock *et al.*, 2010). FD indices have been used more often in evaluation of habitat change effects because they are more sensitive to local habitat quality and landscape structure and are more relevant to ecosystem services than

species diversity indices (Spake *et al.*, 2016; Baulechner *et al.*, 2019).

Conclusions

Our results highlight that overall carabid activity density, species diversity and FD respond differently to both local stand characteristics and landscape structure of surrounding vegetation, so large-scale forest management for conserving carabid diversity must focus on whole assemblages. Significantly, we found that carabid FD, a potentially important factor in evaluating the impact of forest management strategies, can be promoted by relatively simple manipulation for proportions of surrounding vegetation types, since this can be achieved easily and cost-effectively by increasing clusters of grasslands. The weak response of carabid FD to vegetation diversity suggests that any change in landscape composition may be insufficient to result in a significant increase in carabid FD. Our investigation of the drivers of carabid FD amongst natural riparian forest stands in Northwest China has shown that canopy cover is an important stand characteristic of carabid FD, with increasing canopy cover tending to reduce FD. This suggests that management efforts that promote carabid FD should make canopy cover less continuous by creating canopy gaps. Therefore, any management intervention for maximising diversity of ground beetles in forests should consider the pervasive effects of landscape composition and configuration besides local-scale stand factors.

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Conflicts of interest

The authors declare that this article content has none conflict of interest.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1 Identity, abundance, and functional traits of ground beetle individuals collected from 40 sampling plots in natural poplar forests along the Irtysh River in Xinjiang (Northwest China).

Table S2: Spearman's correlation coefficients between landscape factors (proportions of low vegetation productivity (Low VP), medium-low vegetation productivity (Medium-low VP), medium vegetation productivity (Medium VP), medium-high vegetation productivity (Medium-high VP), high vegetation productivity (High VP), overall diversity of vegetation productivity classes (SHDI)) and stand characteristics (canopy cover, vegetation cover, species richness of herbaceous plant species (SR of herbs), average height of herbaceous plant species (Height of herbs) and the distance to river (Dis. to river)). Strong correlations (correlation coefficients $> |0.7|$) are highlighted in bold.

Table S3: Summary of the top-ranked models for estimating responses of activity density, species diversity and functional diversity of ground beetles to the overall diversity of vegetation productivity classes (SHDI), the proportions of medium vegetation productivity (Medium VP), medium-high vegetation productivity (Medium-high VP), high vegetation productivity (High VP), canopy cover, vegetation cover and the nearest distance from sampling plot to river (Dis. to river).

References

- Alalouni, U., Schädler, M. & Brandl, R. (2013) Natural enemies and environmental factors affecting the population dynamics of the gypsy moth. *Journal of Applied Entomology*, **137**, 721–738. <https://doi.org/10.1111/jen.12072>.
- Alberti, M. (2015) Eco-evolutionary dynamics in an urbanizing planet. *Trends in Ecology & Evolution*, **30**, 114–126. <https://doi.org/10.1016/j.tree.2014.11.007>.
- Anton, C., Young, J., Harrison, P.A., Musche, M., Bela, G., Feld, C.K., Harrington, R., Haslett, J.R., Pataki, G., Rounsevell, M.D.A., Skourtos, M., Sousa, J.P., Sykes, M.T., Tinch, R., Vandewalle, M., Watt, A. & Settele, J. (2010) Research needs for incorporating the ecosystem service approach into EU biodiversity conservation policy. *Biodiversity and Conservation*, **19**, 2979–2994. <https://doi.org/10.1007/s10531-010-9853-6>.
- Araujo Calçada, E., Lenoir, J., Plue, J., Broeckx, L.S., Closset-Kopp, D., Hermy, M. & Decocq, G. (2015) Spatial patterns of water-deposited seeds control plant species richness and composition in riparian forest landscapes. *Landscape Ecology*, **30**, 2133–2146. <https://doi.org/10.1007/s10980-015-0236-y>.
- Aubin, I., Venier, L., Pearce, J. & Moretti, M. (2013) Can a trait-based multi-taxa approach improve our assessment of forest management impact on biodiversity? *Biodiversity and Conservation*, **22**, 2957–2975. <https://doi.org/10.1007/s10531-013-0565-6>.
- Balzan, M.V., Bocci, G. & Moonen, A.-C. (2016) Landscape complexity and field margin vegetation diversity enhance natural enemies and reduce herbivory by Lepidoptera pests on tomato crop. *BioControl*, **61**, 141–154. <https://doi.org/10.1007/s10526-015-9711-2>.
- Barsoum, N., Fuller, L., Ashwood, F., Reed, K., Bonnet-Lebrun, A.-S. & Leung, F. (2014) Ground-dwelling spider (Araneae) and carabid beetle (Coleoptera: Carabidae) community assemblages in mixed and monoculture stands of oak (*Quercus robur* L./*Quercus petraea* (Matt.) Liebl.) and Scots pine (*Pinus sylvestris* L.). *Forest Ecology and Management*, **321**, 29–41. <https://doi.org/10.1016/j.foreco.2013.08.063>.

- Bartoń, K. (2018) MuMIn: Multi-Model Inference. R package version 1.40.4. <<https://CRAN.R-project.org/package=MuMIn>>. [accessed on 30 January 2018].
- Batáry, P., Kovács, A. & Báldi, A. (2008) Management effects on carabid beetles and spiders in Central Hungarian grasslands and cereal fields. *Community Ecology*, **9**, 247–254. <https://doi.org/10.1556/ComEc.9.2008.2.14>.
- Bates, D., Maechler, M., Bolker, B. & Walker, S. (2015) Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, **67**, 1–48. <https://doi.org/10.18637/jss.v067.i01>.
- Baulechner, D., Diekötter, T., Wolters, V. & Jauker, F. (2019) Converting arable land into flowering fields changes functional and phylogenetic community structure in ground beetles. *Biological Conservation*, **231**, 51–58. <https://doi.org/10.1016/j.biocon.2019.01.005>.
- Benton, T.G., Vickery, J.A. & Wilson, J.D. (2003) Farmland biodiversity: is habitat heterogeneity the key? *Trends in Ecology & Evolution*, **18**, 182–188. [https://doi.org/10.1016/S0169-5347\(03\)00011-9](https://doi.org/10.1016/S0169-5347(03)00011-9).
- Berlov, O.E. (2002) Atlas of Ground Beetles of Russia. <<https://www.zin.ru/Animalia/Coleoptera/eng/atlasr.htm>>. [accessed on 10 October 2019].
- Bivand, R. (2018) spdep: Spatial Dependence: Weighting Schemes, Statistics. R package version 0.8–1. <<https://CRAN.R-project.org/package=spdep>>. [accessed on 21 November 2018].
- Bodin, Ö. & Norberg, J. (2007) A network approach for analyzing spatially structured populations in fragmented landscape. *Landscape Ecology*, **22**, 31–44. <https://doi.org/10.1007/s10980-006-9015-0>.
- Borchard, F., Buchholz, S., Helbing, F. & Fartmann, T. (2014) Carabid beetles and spiders as bioindicators for the evaluation of montane heathland restoration on former spruce forests. *Biological Conservation*, **178**, 185–192. <https://doi.org/10.1016/j.biocon.2014.08.006>.
- Brambilla, M. & Gobbi, M. (2014) A century of chasing the ice: delayed colonisation of ice-free sites by ground beetles along glacier forelands in the Alps. *Ecography*, **37**, 33–42. <https://doi.org/10.1111/j.1600-0587.2013.00263.x>.
- Chaplin-Kramer, R., O'Rourke, M.E., Blitzer, E.J. & Kremen, C. (2011) A meta-analysis of crop pest and natural enemy response to landscape complexity. *Ecology Letters*, **14**, 922–932. <https://doi.org/10.1111/j.1461-0248.2011.01642.x>.
- Dawson, T.P., Jackson, S.T., House, J.I., Prentice, I.C. & Mace, G.M. (2011) Beyond predictions: biodiversity conservation in a changing climate. *Science*, **332**, 53–58. <https://doi.org/10.1126/science.1200303>.
- Davidowitz, G. & Rosenzweig, M.L. (1998) The latitudinal gradient of species diversity among North American grasshoppers (Acrididae) within a single habitat: a test of the spatial heterogeneity hypothesis. *Journal of Biogeography*, **25**, 553–560. <https://doi.org/10.1046/j.1365-2699.1998.2530553.x>.
- Dormann, C.F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carré, G., Marquéz, J.R.G., Gruber, B., Lafourcade, B., Leitão, P.J., Münkemüller, T., McClean, C., Osborne, P.E., Reineking, B., Schröder, B., Skidmore, A.K., Zurell, D. & Lautenbach, S. (2013) Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. *Ecography*, **36**, 27–46. <https://doi.org/10.1111/j.1600-0587.2012.07348.x>.
- Dudko, R.Y., Matalin, A.V. & Fedorenko, D.N. (2010) The ground beetle fauna (Coleoptera, Carabidae) of Southeastern Altai. *Entomological Review*, **9**, 968–988. <https://doi.org/10.1134/S0013873810080026>.
- Fountain-Jones, N., Baker, S.C. & Jordan, G.J. (2015) Moving beyond the guild concept: developing a practical functional trait framework for terrestrial beetles. *Ecological Entomology*, **40**, 1–13. <https://doi.org/10.1111/een.12158>.
- Fuller, R.L., Oliver, T.H. & Leather, S.R. (2008) Forest management effects on carabid beetle communities in coniferous and broadleaved forests: implications for conservation. *Insect Conservation and Diversity*, **1**, 242–252. <https://doi.org/10.1111/j.1752-4598.2008.00032.x>.
- Fusser, M.S., Pfister, S.C., Entling, M.H. & Schirmel, J. (2017) Effects of field margin type and landscape composition on predatory carabids and slugs in wheat fields. *Agriculture, Ecosystems & Environment*, **247**, 182–188. <https://doi.org/10.1016/j.agee.2017.06.030>.
- Gallé, R., Happe, A.-K., Baillod, A.B., Tscharrntke, T. & Batáry, P. (2019) Landscape configuration, organic management, and within-field position drive functional diversity of spiders and carabids. *Journal of Applied Ecology*, **56**, 63–72. <https://doi.org/10.1111/1365-2664.13257>.
- Gardiner, M.M., Landis, D.A., Gratton, C., DiFonzo, C.D., O'Neal, M., Chacon, J.M., Wayo, M.T., Schmidt, N.P., Mueller, E.E. & Heimpel, G.E. (2009) Landscape diversity enhances biological control of an introduced crop pest in the north-central USA. *Ecological Applications*, **19**, 143–154. <https://doi.org/10.1890/07-1265.1>.
- Gelman, G. & Su, Y.S. (2018) arm: data analysis using regression and multilevel/hierarchical models. R package version 1.10–1. <<https://CRAN.R-project.org/package=arm>>. [accessed on 13 April 2018].
- Gobbi, M. & Fontaneto, D. (2008) Biodiversity of ground beetles (Coleoptera: Carabida) in different habitats of the Italian Po lowland. *Agriculture, Ecosystems & Environment*, **127**, 273–276. <https://doi.org/10.1016/j.agee.2008.04.011>.
- Gobbi, M., Rossaro, B., Vater, A., De Bernardi, F., Pelfini, M. & Brandmayr, P. (2007) Environmental features influencing carabid beetle (Coleoptera) assemblages along a recently deglaciated area in the Alpine region. *Ecological Entomology*, **32**, 682–689. <https://doi.org/10.1111/j.1365-2311.2007.00912.x>.
- Grueber, C.E., Nakagawa, S., Laws, R.J. & Jamieson, I.G. (2011) Multi-model inference in ecology and evolution: challenges and solutions. *Journal of Evolutionary Biology*, **24**, 699–711. <https://doi.org/10.1111/j.1420-9101.2010.02210.x>.
- Homburg, K., Homburg, N., Schäfer, F., Schuldt, A. & Assmann, T. (2014) Carabids.org - a dynamic online database of ground beetle species traits (Coleoptera, Carabidae). *Insect Conservation and Diversity*, **7**, 195–205. <https://doi.org/10.1111/icad.12045>.
- Humphrey, J.W., Hawes, C., Peace, A.J., Ferris-Kaan, R. & Jukes, M.R. (1999) Relationships between insect diversity and habitat characteristics in plantation forests. *Forest Ecology and Management*, **113**, 11–21. [https://doi.org/10.1016/S0378-1127\(98\)00413-7](https://doi.org/10.1016/S0378-1127(98)00413-7).
- Jackson, H.B. & Fahrig, L. (2012) What size is a biologically relevant landscape? *Landscape Ecology*, **27**, 929–941. <https://doi.org/10.1007/s10980-012-9757-9>.
- Jelaska, L.Š. & Durbešić, P. (2009) Comparison of the body size and wing form of carabid species (Coleoptera: Carabidae) between isolated and continuous forest habitats. *Annales de la Société entomologique de France (N.S.)*, **45**, 327–338. <https://doi.org/10.1080/00379271.2009.10697618>.
- Kataev, B.M. & Liang, H.B. (2015) Taxonomic review of Chinese species of ground beetles of the subgenus *Pseudophonus* (genus *Harpalus*) (Coleoptera: Carabidae). *Zootaxa*, **3930**, 1–39. <https://doi.org/10.11646/zootaxa.3920.1.1>.
- Kleyer, M., Dray, S., de Bello, F., Lepš, J., Pakeman, R.J., Strauss, B., Thuiller, W. & Lavorel, S. (2012) Assessing species and community functional responses to environmental gradients: which multivariate methods? *Journal of Vegetation Science*, **23**, 805–821. <https://doi.org/10.1111/j.1654-1103.2012.01402.x>.
- Koivula, M., Kukkonen, J. & Niemä, J. (2002) Boreal carabid-beetle (Coleoptera, Carabidae) assemblages along the clear-cut originated succession gradient. *Biodiversity and Conservation*, **11**, 1269–1288. <https://doi.org/10.1023/A:1016018702894>.

- Koivula, M.J. & Vermeulen, H.J.W. (2005) Highways and forest fragmentation - effects on carabid beetles (Coleoptera, Carabidae). *Landscape Ecology*, **20**, 911–926. <https://doi.org/10.1007/s10980-005-7301-x>.
- Lafage, D., Maugeness, S., Bouzillé, J.-B. & Pétilion, J. (2015) Disentangling the influence of local and landscape factors on alpha and beta diversities: opposite response of plants and ground-dwelling arthropods in wet meadows. *Ecological Research*, **30**, 1025–1035. <https://doi.org/10.1007/s11284-015-1304-0>.
- Lafage, D., Secondi, J., Georges, A., Bouzillé, J.-B. & Pétilion, J. (2014) Satellite-derived vegetation indices as surrogate of species richness and abundance of ground beetles in temperate floodplains. *Insect Conservation and Diversity*, **7**, 327–333. <https://doi.org/10.1111/icad.12056>.
- Lange, M., Türke, M., Pašalić, E., Boch, S., Hessenmöller, D., Müller, J., Prati, D., Socher, S.A., Fischer, M., Weisser, W.W. & Gossner, M.M. (2014) Effects of forest management on ground-dwelling beetles (Coleoptera; Carabidae, Staphylinidae) in Central Europe are mainly mediated by changes in forest structure. *Forest Ecology and Management*, **329**, 166–176. <https://doi.org/10.1016/j.foreco.2014.06.012>.
- Lavallee, S.L. & Richardson, J.S. (2010) Relative abundance and movement of the carabid beetle *Scaphinotus angusticollis* in managed coniferous riparian forests of southwestern British Columbia. *Canadian Journal of Forest Research*, **40**, 611–618. <https://doi.org/10.1139/x10-003>.
- LeVine, D. & Crews, K. (2019) Time series harmonic regression analysis reveals seasonal vegetation productivity trends in semi-arid savannas. *International Journal of Applied Earth Observation and Geoinformation*, **80**, 94–101. <https://doi.org/10.1016/j.jag.2019.04.007>.
- Lezama, F., Baeza, S., Altesor, A., Cesa, A., Chaneton, E.J. & Paruelo, J. M. (2014) Variation of grazing-induced vegetation changes across a large-scale productivity gradient. *Journal of Vegetation Science*, **25**, 8–21. <https://doi.org/10.1111/jvs.12053>.
- Lin, S.Z., Zhang, Z.Y., Zhang, Q. & Lin, Y.Z. (2006) Progress in the study of molecular genetic improvements of poplar in China. *Journal of Integrative Plant Biology*, **48**, 1001–1007. <https://doi.org/10.1111/j.1744-7909.2006.00303.x>.
- Macfadyen, S., Kramer, E.A., Parry, H.R. & Schellhorn, N.A. (2015) Temporal change in vegetation productivity in grain production landscapes: linking landscape complexity with pest and natural enemy communities. *Ecological Entomology*, **40**, 56–69. <https://doi.org/10.1111/een.12213>.
- Magagula, C.N. (2003) Changes in carabid beetle diversity within a fragmented agricultural landscape. *African Journal of Ecology*, **41**, 23–30. <https://doi.org/10.1046/j.1365-2028.2003.00403.x>.
- Magura, T. (2017) Ignoring functional and phylogenetic features masks the edge influence on ground beetle diversity across forest-grassland gradient. *Forest Ecology and Management*, **384**, 371–377. <https://doi.org/10.1016/j.foreco.2016.10.056>.
- Nakagawa, S. & Schielzeth, H. (2013) A general and simple method for obtaining R^2 from generalized linear mixed-effects models. *Methods in Ecology and Evolution*, **4**, 133–142. <https://doi.org/10.1111/j.2041-210x.2012.00261.x>.
- Negro, M., Vacchiano, G., Berretti, R., Chamberlain, D.E., Palestini, C., Motta, R. & Rolando, A. (2014) Effects of forest management on ground beetle diversity in alpine beech (*Fagus sylvatica* L.) stands. *Forest Ecology and Management*, **328**, 300–309. <https://doi.org/10.1016/j.foreco.2014.05.049>.
- Niemelä, J. & Kotze, D.J. (2009) Carabid beetle assemblages along urban to rural gradients: a review. *Landscape and Urban Planning*, **92**, 65–71. <https://doi.org/10.1016/j.landurbplan.2009.05.016>.
- Niemelä, J., Kotze, D.J., Venn, S., Penev, L., Stoyanov, I., Spence, J., Hartley, D. & de Oca, E.M. (2002) Carabid beetle assemblages (Coleoptera, Carabidae) across urban-rural gradients: an international comparison. *Landscape Ecology*, **17**, 387–401. <https://doi.org/10.1023/A:1021270121630>.
- Oiver, T., Roy, D.B., Hill, J.K., Brerethon, T. & Thomas, C.D. (2010) Heterogeneous landscapes promote population stability. *Ecology Letters*, **13**, 473–484. <https://doi.org/10.1111/j.1461-0248.2010.01441.x>.
- Oksanen, J., Blanchet, F.G., Friendly, M., Kindt, R., Legendre, P., McGinn, D., Minchin, P.R., O'Hara, R.B., Simpson, G.L., Solymos, P., Henry, M., Stevens, H., Szoecs, E. & Wagner, H. (2019) vegan: community Ecology Package. R package version 2.5-4. <<https://CRAN.R-project.org/package=vegan>>. [accessed on 4 February 2019].
- Pavoine, S. & Bonsall, M.B. (2011) Measuring biodiversity to explain community assembly: a unified approach. *Biological Reviews*, **86**, 792–812. <https://doi.org/10.1111/j.1469-185X.2010.00171.x>.
- Petchey, O.L. & Gaston, K.J. (2002) Functional diversity (FD), species richness and community composition. *Ecology Letters*, **5**, 402–411. <https://doi.org/10.1046/j.1461-0248.2002.00339.x>.
- Petchey, O.L. & Gaston, K.J. (2006) Functional diversity: back to basics and looking forward. *Ecology Letters*, **9**, 741–758. <https://doi.org/10.1111/j.1461-0248.2006.00924.x>.
- Philippe, R.N. & Bohlmann, J. (2007) Poplar defense against insect herbivores. *Canadian Journal of Botany*, **85**, 1111–1126. <https://doi.org/10.1139/b07-109>.
- Philpott, S.M., Albuquerque, S., Bichier, P., Cohen, H., Egerer, M.H., Kirk, C. & Will, K.W. (2019) Local and landscape drivers of carabid activity, species richness, and traits in urban gardens in coastal California. *Insects*, **10**, 112. <https://doi.org/10.3390/insects10040112>.
- Purtauf, T., Dauber, J. & Wolters, V. (2005) The response of carabids to landscape simplification differs between trophic groups. *Oecologia*, **142**, 458–464. <https://doi.org/10.1007/s00442-004-1740-y>.
- Rahman, M.L., Tarrant, S., McCollin, D. & Ollerton, J. (2015) Vegetation cover and grasslands in the vicinity accelerate development of carabid beetle assemblages on restored landfill sites. *Zoology and Ecology*, **25**, 347–354. <https://doi.org/10.1080/21658005.2015.1068521>.
- Rainio, J. & Niemelä, J. (2003) Ground beetles (Coleoptera: Carabidae) as bioindicators. *Biodiversity and Conservation*, **12**, 487–506. <https://doi.org/10.1023/A:1022412617568>.
- Rao, C.R. (1982) Diversity and dissimilarity coefficients: unified approach. *Theoretical Population Biology*, **21**, 24–43. [https://doi.org/10.1016/0040-5809\(82\)90004-1](https://doi.org/10.1016/0040-5809(82)90004-1).
- R Core Team (2018) *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria. <https://www.r-project.org/>.
- Ricotta, C. (2005) A note on functional diversity measures. *Basic and Applied Ecology*, **6**, 479–486. <https://doi.org/10.1016/j.baae.2005.02.008>.
- Rooney, N., McCann, K.S. & Moore, J.C. (2008) A landscape theory for food web architecture. *Ecology Letters*, **11**, 867–881. <https://doi.org/10.1111/j.1461-0248.2008.01193.x>.
- Running, S.W., Nemani, R.R., Heinsch, F.A., Zhao, M.S., Reeves, M. & Hashimoto, H. (2004) A continuous satellite-derived measure of global terrestrial primary production. *BioScience*, **54**, 547–560. [https://doi.org/10.1641/0006-3568\(2004\)054\[0547:ACSMOG\]2.0.CO;2](https://doi.org/10.1641/0006-3568(2004)054[0547:ACSMOG]2.0.CO;2).
- Ryken, J.J., Moldenke, A.R. & Olson, D.H. (2007) Headwater riparian forest-floor invertebrate communities associated with alternative forest management practices. *Ecological Applications*, **17**, 1168–1183. <https://doi.org/10.1890/06-0901>.
- Santos, P.Z.F., Crouzeilles, R. & Sansevero, J.B.B. (2019) Can agroforestry systems enhance biodiversity and ecosystem service provision in agricultural landscapes? A meta-analysis for the Brazilian Atlantic

- Forest. *Forest Ecology and Management*, **433**, 140–145. <https://doi.org/10.1016/j.foreco.2018.10.064>.
- Schirmel, J., Albrecht, M., Bauer, P.-M., Sutter, L., Pfister, S.C. & Entling, M.H. (2018) Landscape complexity promotes hoverflies across different types of semi-natural habitats in farmland. *Journal of Applied Ecology*, **55**, 1747–1758. <https://doi.org/10.1111/1365-2664.13095>.
- Schlegel, J. & Riesen, M. (2012) Environmental gradients and succession patterns of carabid beetles (Coleoptera: Carabidae) in an Alpine glacier retreat zone. *Journal of Insect Conservation*, **16**, 657–675. <https://doi.org/10.1007/s10841-011-9448-x>.
- Scrucca, L. (2014) qcc: Quality Control Charts. R package version 2.6. <<https://CRAN.R-project.org/package=qcc>>. [accessed on 15 August 2019].
- Spake, R., Barsoum, N., Newton, A.C. & Doncaster, C.P. (2016) Drivers of the composition and diversity of carabid functional traits in UK coniferous plantations. *Forest Ecology and Management*, **359**, 300–308. <https://doi.org/10.1016/j.foreco.2015.10.008>.
- Straub, C.S., Finke, D.L. & Snyder, W.E. (2008) Are the conservation of natural enemy biodiversity and biological control compatible goals? *Biological Control*, **45**, 225–237. <https://doi.org/10.1016/j.biocontrol.2007.05.013>.
- Taboada, A., Kotze, D.J., Tárrega, R. & Salgado, J.M. (2006) Traditional forest management: do carabid beetles respond to human-created vegetation structures in an oak mosaic landscape? *Forest Ecology and Management*, **237**, 436–449. <https://doi.org/10.1016/j.foreco.2006.09.077>.
- Tews, J., Brose, U., Grimm, V., Tielbörger, K., Wichmann, M.C., Schwager, M. & Jeltsch, F. (2004) Animal species diversity driven by habitat heterogeneity/diversity: the importance of keystone structures. *Journal of Biogeography*, **31**, 79–92. <https://doi.org/10.1046/j.0305-0270.2003.00994.x>.
- Tscharntke, T., Bommarco, R., Clough, Y., Crist, T.O., Kleijn, D., Rand, T.A., Tylianakis, J.M., van Nouhuys, S. & Vidal, S. (2007) Conservation biological control and enemy diversity on a landscape scale. *Biological Control*, **43**, 294–309. <https://doi.org/10.1016/j.biocontrol.2007.08.006>.
- Vasseur, C., Joannon, A., Aviron, S., Burel, F., Meynard, J.M. & Baudry, J. (2013) The cropping systems mosaic: how does the hidden heterogeneity of agricultural landscapes drive arthropod populations? *Agriculture, Ecosystems & Environment*, **166**, 3–14. <https://doi.org/10.1016/j.agee.2012.08.013>.
- Vele, A., Holusa, J., Frouz, J. & Konvicka, O. (2011) Local and landscape drivers of ant and carabid beetle communities during spruce forest succession. *European Journal of Soil Biology*, **47**, 349–356. <https://doi.org/10.1016/j.ejsobi.2011.09.003>.
- Voepel, H., Ruddell, B., Schumer, R., Troch, P.A., Brooks, P.D., Neal, A., Durcik, M. & Sivapalan, M. (2011) Quantifying the role of climate and landscape characteristics on hydrologic partitioning and vegetation response. *Water Resources Research*, **47**, 427–438. <https://doi.org/10.1029/2010wr009944>.
- Wang, B.L., Tian, C.M. & Sun, J.L. (2019) Effects of landscape complexity and stand factors on arthropod communities in poplar forests. *Ecology and Evolution*, **9**, 7143–7156. <https://doi.org/10.1002/ece3.528>.
- Weiskittel, A.R., Crookston, N.L. & Radtke, P.J. (2011) Linking climate, gross primary productivity, and site index across forests of the western United States. *Canadian Journal of Forest Research*, **41**, 1710–1721. <https://doi.org/10.1139/x11-086>.
- Werling, B.P. & Gratton, C. (2008) Influence of field margins and landscape context on ground beetle diversity in Wisconsin (USA) potato fields. *Agriculture, Ecosystems & Environment*, **128**, 104–108. <https://doi.org/10.1016/j.agee.2008.05.007>.
- Woodcock, B.A., Harrower, C., Redhead, J., Edwards, M., Vanbergen, A. J., Heard, M.S., Roy, D.B. & Pywell, R.F. (2014) National patterns of functional diversity and redundancy in predatory ground beetles and bees associated with key UK arable crops. *Journal of Applied Ecology*, **51**, 142–151. <https://doi.org/10.1111/1365-2664.12171>.
- Woodcock, B.A., Leather, S.R. & Watt, A.D. (2003) Changing management in Scottish birch woodlands: a potential threat to local invertebrate biodiversity. *Bulletin of Entomological Research*, **93**, 159–167. <https://doi.org/10.1079/BER2003227>.
- Woodcock, B.A., Redhead, J., Vanbergen, A.J., Hulmes, L., Hulmes, S., Peyton, J., Nowakowski, M., Pywell, R.F. & Heard, M.S. (2010) Impact of habitat type and landscape structure on biomass, species richness and functional diversity of ground beetles. *Agriculture, Ecosystems & Environment*, **139**, 181–186. <https://doi.org/10.1016/j.agee.2010.07.018>.
- Woodcock, B.A., Westbury, D.B., Potts, S.G., Harris, S.J. & Brown, V. K. (2005) Establishing field margins to promote beetle conservation in arable farms. *Agriculture, Ecosystems & Environment*, **107**, 255–266. <https://doi.org/10.1016/j.agee.2004.10.029>.
- Yang, L., Zhang, Q., Liu, B., Zeng, Y.D., Pan, Y.F., Li, M.L. & Lu, Y.H. (2019) Mixed effects of landscape complexity and insecticide use on ladybeetle abundance in wheat fields. *Pest Management Science*, **75**, 1638–1645. <https://doi.org/10.1002/ps.5281>.
- Zuur, A.F., Ieno, E.N., Walker, N.J., Saveliev, A.A. & Smith, G.M. (2009) Generalised estimation equations. *Mixed effects models and extensions in ecology with R*. Statistics for Biology and Health, pp. 295–321. Springer, New York, NY. https://doi.org/10.1007/978-0-387-87458-6_12.

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